The Collected Mathematical Papers of Arthur Cayley, Vol. VI

Pages 101–150 (Papers 396–402)

Transcription

396. On a Certain Envelope Depending on a Triangle Inscribed in a Circle

[From the Philosophical Magazine, vol. xxxiv. (1867), pp. 174–179.]

[Continued from p. 73.]

[Pp. 175—179.]

It is interesting to verify this; I write $Z = -X - Y$, the cubic function then assumes the form

$$(a, \beta, \gamma, \delta X, Y)^3,$$

where $(a, \beta, \gamma, \delta)$ have the values presently given.

The Hessian is

$$(2a\gamma - 2\beta^2)X^2 + (a\delta - \beta\gamma) \cdot 2XY + (2\beta\delta - 2\gamma^2)Y^2,$$

or writing $2XY = Z^2 - X^2 - Y^2$, this is

$$= (2a\gamma - 2\beta^2 - a\delta + \beta\gamma)X^2 + (2\beta\delta - 2\gamma^2 - a\delta - \beta\gamma)Y^2 + (a\delta - \beta\gamma)Z^2.$$

We find after some easy reductions,

$$\begin{aligned} J_0 &= Z^2 + L'^2, = -3(a + c)(ac + M), \\ I' &= -H' + 2K' + I' + F', = 3a(a + c)(b + c), \\ \gamma &= -K' + 2H' + J' + G', = 3b(a + c)(b + c), \\ \delta &= I' + J', = -3(b + c)(bc + M), \end{aligned}$$

and hence

$$a\delta - \beta\gamma = 81(a + c)(b + c)\{(ac + M)(bc + M) - ab(a + c)(b + c)\},$$

where the expression in {} is

$$\begin{aligned} &= (ab + 2ac + bc)(ab + ac + 2bc) - ab(ab + ac + bc + c^2), \\ &= c\{ab(3a + 3b) + c(b + 2a)(a + 2b) - ab(a + b + c)\}, \\ &= 2c(a + b)(bc + ca + ab), \end{aligned}$$

and therefore

$$a\delta - \beta\gamma = -162(b + c)(c + a)(a + b)M;$$

the other coefficients may be similarly calculated, and omitting the merely numerical factor, we have

$$\text{Hess} = (b + c)(c + a)(a + b)M(aX^2 + bY^2 + cZ^2),$$

which is right.

I write next

$$\begin{aligned} HU + 3MU - \psi U = X^2(-a^2x + I'Y - F'Z) \\ + Y^2(-b^2y + J'Z - G'X) + Z^2(-a^2z + K'X - H'Y), \end{aligned}$$

or writing $x = z - Y$, $y = z + X$, this is

$$HU + 3MU - \psi U = -4^2 z(aX^2 + bY^2 + cZ^2) \\ + X^2\{(I' + a^2)Y - F'Z\} + Y^2\{J'Z - (G' + b^2)X\} + Z^2\{K'X - H'Y\},$$

we may determine ψ, so that the cubic function of X, Y, Z contains the factor $aX^2 + bY^2 + cZ^2$; writing $Z = -X - Y$, then

Contains the factor $(a + c)X^2 + 2cXY + (b + c)Y^2$

Quotient is $-3(ac + M)X + 3(bc + M)Y$.

$$Z^2(\lambda I'' + F') + Z^2Y(-H' + 2K' + I' + \tfrac{1}{4}\psi - a^2) \\ + ZY^2(K' - 2H' - G' - J' - b^2 + \psi) + Y^3(-J' - H').$$

We have seen that

$$K' + F' = -3(a + c)(ac + M),$$
$$J' + H' = -3(b + c)(bc + M),$$

whence the quotient is, as above stated,

$$= -3(ac + M)X + 3(bc + M)Y.$$

Comparing the coefficients of X^2Y, we have

$$\begin{aligned} a\psi &= -(-F' + 2K' + I' + I') + 3(a + c)(bc + M) - 6c(ac + M), \\ &= 9a(a + c)(b + c) + 3(a + c)(ab + ac + 2bc) - 6c(ab + 2ac + bc), \\ &= 12a(bc + ca + ab) = 12aM, \end{aligned}$$

that is $\psi = 12M$; and the same value would have been obtained by comparing the coefficients of XY^2. Hence $HU + 3MU - \psi U$ divides by $aX^2 + bY^2 + cZ^2$, the quotient being

$$\begin{aligned} &= -12Mz - 3(ac + M)X + 3(bc + M)Y, \\ &= -12Mz - 3(ac + M)(y - z) + 3(bc + M)(z - x), \\ &= -3\{(bc + M)x + (ca + M)y + (ab + M)z\}, \end{aligned}$$

which is or, finally it is

$$= -3\{(bc + M)x + (ca + M)y + (ab + M)z\},$$

and we thus have

$$HU + 3MU - \psi U = -3(aX^2 + bY^2 + cZ^2) \times \{(bc + M)x + (ca + M)y + (ab + M)z\},$$

so that the three inflexions are the intersections of the cubic curve by the line

$$(bc + M)x + (ca + M)y + (ab + M)z = 0.$$

It may be noticed, that if we write

$$x + y + z = u,$$
$$bex + cay + abz = -Mu,$$
$$ax + by + cz = v,$$

then x, y, z will be as

$$(b - c)\{(2M - bc)u - av\} : (c - a)\{(2M - ca)u - bv\} : (a - b)\{(2M - ab)u - cv\},$$

and substituting these values in the equation

$$ax(y-z)^2 + by(z-x)^2 + cz(x-y)^2 = 0$$

of the cubic, we have a cubic equation for the ratio $(u : v)$; and thence the values (x, y, z) for the coordinates of the inflexions.

It may be added, that we have

$$\begin{aligned}12MU = -3(aX^2 + bY^2 + cZ^2)\{(bc + M)x + (ca + M)y + (ab + M)z\} \\ + \{X^2(FY - F'Z) + Y^2(J'Z - G'X) + Z^2(K'X - H'Y)\} = 0,\end{aligned}$$

which is the equation of the cubic expressed in the canonical form.

[Pp. 175—179.]

Effecting the process indicated p. 73, but writing for greater convenience (x, y, z) in place of (X, Y, Z), so that the substitution to be made is

$$(a, b, c, f, g, h, i, j, k, l) = (6x, 6y, 6z, -2z, -2x, -2z, -2x, -2y, x + y + z),$$

respectively (where I have corrected a misprint in the formula as originally given) I find the equation of the envelope to be

$$\begin{aligned}
&4y^2(y-z)^2a^4 + 4z^2x(z-x)^2b^4 + 4x^2y(x-y)^2c^4 \\
&\quad + 4zx(z^2 + x^2 + 3yz - 2zx + 5xy)b^2c \\
&\quad + 4xy(x^2 + y^2 + 3zx - 2xy + 5yz)c^2a \\
&\quad + 4yz(y^2 + z^2 + 3xy - 2yz + 5zx)a^2b \\
&\quad + 4xy(x^2 + y^2 + 3yz - 2xy + 5zx)bc^2 \\
&\quad + 4y^2(y^2 + z^2 + 3zx - 2yz + 5xy)ca^2 \\
&\quad + 4zx(z^2 + x^2 + 3xy - 2zx + 5yz)ab^2 \\
&\quad + x(x^3 - 2x^2y - 2x^2z + xy^2 + 33xyz + xz^2 + 12y^2z + 12yz^2)b^2c^2 \\
&\quad + y(y^3 - 2y^2z - 2y^2x + yz^2 + 33xyz + yx^2 + 12z^2x + 12zx^2)c^2a^2 \\
&\quad + z(z^3 - 2z^2x - 2z^2y + zx^2 + 33xyz + zy^2 + 12x^2y + 12xy^2)a^2b^2 \\
&\quad + 2yz(11x^2 + y^2 + z^2 - 2yz + 24xy + 24zx)a^2bc \\
&\quad + 2zx(11y^2 + z^2 + x^2 - 2zx + 24yz + 24xy)b^2ca \\
&\quad + 2xy(11z^2 + x^2 + y^2 - 2xy + 24zx + 24yz)c^2ab = 0.
\end{aligned}$$

The function on the left-hand side is the quotient by $-48(a + b + c)^2$ of the sextic function, Table 67 of my third Memoir on Quantics, [144]; the foregoing quotient was calculated without using the coefficient of the term in $a^2b^2c^2$ (viz.?") of the table, but by way of verification, I calculated from the table the term in question, and found it to be

$$\begin{aligned}&(x + y + z)^4 - 2(x + y + z)^2(yz + zx + xy) \\ &\qquad + 296(x + y + z)xyz - 8(y^2z^2 + z^2x^2 + x^2y^2),\end{aligned}$$

and this should consequently be equal to the coefficient of $a^2b^2c^2$ in the product of $(a + b + c)^2$ into the foregoing quartic function of (a, b, c) that is, it should be

$$\begin{aligned}
&= x(x^3 - 2x^2y - 2x^2z + xy^2 + 33xyz + xz^2 + 12y^2z + 12yz^2) \\
&\quad + y(y^3 - 2y^2z - 2y^2x + yz^2 + 33xyz + yx^2 + 12z^2x + 12zx^2) \\
&\quad + z(z^3 - 2z^2x - 2z^2y + zx^2 + 33xyz + zy^2 + 12x^2y + 12xy^2) \\
&\quad + 4yz(11x^2 + y^2 + z^2 - 2yz + 24xy + 24zx) \\
&\quad + 4zx(11y^2 + z^2 + x^2 - 2zx + 24yz + 24xy) \\
&\quad + 4xy(11z^2 + x^2 + y^2 - 2xy + 24zx + 24yz),
\end{aligned}$$

which is accordingly found to be the case.

397. Specimen Table $M \equiv a^\alpha b^\beta$ (Mod. N) for Any Prime or Composite Modulus

[From the Quarterly Journal of Pure and Applied Mathematics, vol. ix. (1868), pp. 95—96 and plate.]

If N be a prime number, and a one of its primitive roots, then any number M prime to N, or what is the same thing, any number in the series $1, 2, \ldots N-1$, may be exhibited in the form $M = a^\alpha$ (Mod. N); where α is said to be the *index* of M in regard to the particular root a.

Jacobi's *Canon Arithmeticus* (Berlin, 1839), contains a series of tables, giving the indices of the numbers $1, 2, 3 \ldots N-1$ for every prime number N less than 1000, and giving conversely for each such prime number the numbers M which correspond to the indices $\alpha = 1, 2, \ldots (N-1)$ (Tabulæ Numerorum ad Indices datos pertinentium et Indicum Numero dato correspondentium).

A similar theory applies, it is well known, to the composite numbers; the only difference is, that in order to exhibit for a given composite number N the different numbers less than N and prime to it, we require not a single root a, but two or more roots $a, b, \ldots$ and that in terms of these we have $M = a^\alpha b^\beta \ldots$ (Mod. N).

For each root a there is an index A (or say the Indicator of the root), such that $a^A = 1$ (Mod. N), A being the least index for which this equation is satisfied; and the indices $\alpha, \beta, \ldots$ extend from 1 to $A, B, \ldots$ respectively; the number of different combinations or the product $AB \ldots$, being precisely equal to $\phi(N)$, the number of integers less than N and prime to it.

The least common multiple of $A, B \ldots$, is termed the Maximum Indicator, and representing it by I, then for any number M not prime to N, we have $M^I = 1$ (Mod. N), a theorem made use of by Cauchy for the solution of indeterminate equations of the first order.

Thus $N = 20$, the roots may be taken to be 3, 11; the corresponding exponents are 4, 2 (viz. $3^4 \equiv 1$ (Mod. 20), $11^2 \equiv 1$ (Mod. 20)), and the product of these is 8, the number of integers less than 20 and prime to it; the series [go to p. 86]

[Plate pages 84–85. Tables of indices for moduli $N = 1$ to 50. The tables occupy two full pages with the following arrangement:]

Table arrangement. For each modulus N from 1 to 50, the table presents five heading lines:

1. The modulus N;
2. The root or roots belonging to each number N;
3. The indicators of these roots;
4. The maximum indicator;
5. The number $\phi(N)$.

The remaining lines contain the index or indices of each of the $\phi(N)$ numbers M less than N and prime to it, the number corresponding to such index or indices being placed outside in the same horizontal line.

Specimen: $N = 30$.

30 has the roots 7, 11, indices 4, 2 respectively; the Maximum Indicator is 4, and the number of integers less than 30 and prime to it is 8; taking any such number, say 17, the indices are 1, 1, that is, we have $17 = 7^1 \cdot 11^1$ (Mod. 30).

[From p. 83]

The foregoing corresponds to the Tabulæ Indicum Numero dato correspondentium of Jacobi; on account of multiplicity of roots there does not appear to be any mode of forming a single table corresponding to the Tabulæ Numerorum ad. Indices datos pertinentium; and there would be no adequate advantage in forming for each number N a separate table in some such form as

$N = 20$

Roots	3	11
Nos.		
1	0	0
3	1	0
7	1	1
9	2	0
11	0	1
13	2	1
17	3	1
19	3	0

which I have written down in the form of a table of single entry; for although (whenever, as in the present case, the number of roots is only two) it might have been better exhibited as a table of double entry, when the number of roots is three or more it could not of course be exhibited as a table of corresponding multiple entry.

398. On a Certain Sextic Developable, and Sextic Surface Connected Therewith

[From the Quarterly Journal of Pure and Applied Mathematics, vol. ix. (1868), pp. 129—142 and 373—376.]

I PROPOSE to consider [first] the sextic developable derived from a quartic equation, viz. taking this to be $(a, b, c, d, et, 1)^4 = 0$, where (a, b, c, d, e) are any linear functions of the coordinates (x, y, z, w), the equation of the developable in question is

$$(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e + 2bcd - c^3)^2 = 0.$$

I have already, in the paper "On a Special Sextic Developable," Quarterly Journal of Mathematics, vol. VII. (1866), pp. 105—113, [373], considered a particular case of this surface, viz. that in which c was $= 0$, the geometrical peculiarity of which is that the cuspidal edge is there an excubo-quartic curve (of a special form, having two stationary tangents), whereas in the general case here considered it is a sextic curve.

There was analytically the convenience that the linear functions being only the four functions a, b, d, e, these could be themselves taken as coordinates, whereas in the present case we have the five linear functions a, b, c, d, e.

The developable $(ae - 4bd + 3c^2)^3 - 27(ace - ad^2 - b^2e + 2bcd - c^3)^2 = 0$ is a sextic developable having for its cuspidal curve the sextic curve

$$ae - 4bd + 3c^2 = 0,$$
$$ace - ad^2 - b^2e + 2bcd - c^3 = 0,$$

(say $I = 0$, $J = 0$, as usual), and having besides a nodal curve the equations of which may be written

$$6(ac - b^2) : 3(ad - bc) : ae + 2bd - 3c^2 : 3(be - cd) : 6(ce - d^2) = a : b : c : d : e.$$

viz. these equations are really equivalent to two equations, and they represent a curve of the fourth order which is an excubo-quartic.

We may in fact find the equations of the nodal curve by assuming $(a, b, c, d, e\between t, 1)^4$ to be a perfect square, say to avoid fractions that it is $= 3(\alpha t^2 + 2\beta t + \gamma)^2$, then we have

$$a : b : c : d : e = 3\alpha^2 : 3\alpha\beta : \alpha\gamma + 2\beta^2 : \beta\gamma : 3\gamma^2,$$

which equations as involving the two arbitrary parameters $\alpha : \beta : \gamma$, give two equations between (a, b, c, d, e), and we may at once by means of them verify the above-mentioned equations of the nodal curve.

It also hereby appears that the nodal curve is as stated an excubo-quartic curve; viz. we have between a, b, c, d, e a single linear relation, that is a quadric relation between α, β, γ, and this equation may be satisfied identically by taking for α, β, γ properly determined quadric functions of a variable parameter θ; whence a, b, c, d, e are proportional to quartic functions of the variable parameter θ, or the curve is an excubo-quartic.

The equations of the nodal curve may be presented under a somewhat different form; viz. the cubi-covariant of $(a, b, c, d, e\between t, 1)^4 = 0$ being

$$\begin{aligned}
(-a^2d + 3abc - 2b^3 \quad & - a^2e - 2abd + 9ac^2 - 6b^2c \\
& - abe + 6acd - 6bc^2 \quad + 3ad^2 - b^2e \\
& + ade + 2bd^2 - 3bce \\
& + ae^2 + 2bde - 9c^2e + 6cd^2 \\
& \qquad\qquad + be^2 - 3cde + 2d^3 \between t, 1)^3 = 0,
\end{aligned}$$

say this function, multiplied by 6 to avoid fractions, is

$$(\mathfrak{a}, \mathfrak{b}, \mathfrak{c}, \mathfrak{d}, \mathfrak{e}, \mathfrak{f}, \mathfrak{g} \between t, 1)^3,$$

that is

$$\begin{aligned}
\mathfrak{a} &= 6(-a^2d + 3abc - 2b^3), \\
\mathfrak{b} &= (-a^2e - 2abd + 9ac^2 - 6b^2c), \\
\mathfrak{c} &= 2(-abe + 6acd - 6bc^2), \\
\mathfrak{d} &= 3(+ad^2 - b^2e), \\
\mathfrak{e} &= 2(+ade + 2bd^2 - 3bce), \\
\mathfrak{f} &= (+ae^2 + 2bde - 9c^2e + 6cd^2), \\
\mathfrak{g} &= 6(+be^2 - 3cde + 2d^3),
\end{aligned}$$

then the equations of the nodal curve may be written:

$$\mathfrak{a} = 0,\ \mathfrak{b} = 0,\ \mathfrak{c} = 0,\ \mathfrak{d} = 0,\ \mathfrak{e} = 0,\ \mathfrak{f} = 0,\ \mathfrak{g} = 0.$$

It may be mentioned that we have identically

$$\begin{aligned}
&ae - 4bd + 3c^2 = 0,\\
&af - 3be + 2cd = 0,\\
&ag - 9ce + 8d^2 = 0,\\
&bg - 3cf + 2de = 0,\\
&bf - 4ce + 3d^2 = eg - 3df + 2e^2 = 0.
\end{aligned}$$

and moreover

$$ag - 6bf + 15ce - 10d^2 = -6(bf - 4ce + 3d^2), = +6(I^2 - 27J^2),$$

so that the equation of the developable may be written in the form

$$ag - 6bf + 15ce - 10d^2 = 0,$$

or in the more simple form

$$bf - 4ce + 3d^2 = 0,$$

each of which puts in evidence the nodal curve on the surface.

The nodal and cuspidal curves meet in the points

$$\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{d}{e},$$

being, as it is easy to show, a system of four points.

The four points in question form a tetrahedron, the equations of the faces of which may be taken to be $x = 0$, $y = 0$, $z = 0$, $w = 0$; and the equation of the surface may be expressed in this system of quadriplanar coordinates.

We introduce these coordinates *ab initio*, by taking the quartic function of t to be

$$(a, b, c, d, et, 1)^4 = x'(t + \alpha)^4 + y'(t + \beta)^4 + z'(t + \gamma)^4 + w'(t + \delta)^4,$$

that is, by writing

$$\begin{aligned}
a &= x' + y' + z' + w',\\
b &= \alpha x' + \beta y' + \gamma z' + \delta w',\\
c &= \alpha^2 x' + \beta^2 y' + \gamma^2 z' + \delta^2 w',\\
d &= \alpha^3 x' + \beta^3 y' + \gamma^3 z' + \delta^3 w',\\
e &= \alpha^4 x' + \beta^4 y' + \gamma^4 z' + \delta^4 w'.
\end{aligned}$$

Observe that (t_1, t_2, t_3, t_4) being any constant quantities, we thence have

$$\begin{aligned}
&e - d\Sigma t_1 + c\Sigma t_1 t_2 - b\Sigma t_1 t_2 t_3 + a t_1 t_2 t_3 t_4\\
&\qquad = x'(\alpha - t_1)(\alpha - t_2)(\alpha - t_3)(\alpha - t_4)\\
&\qquad + y'(\beta - t_1)(\beta - t_2)(\beta - t_3)(\beta - t_4)\\
&\qquad + z'(\gamma - t_1)(\gamma - t_2)(\gamma - t_3)(\gamma - t_4)\\
&\qquad\qquad + w'(\delta - t_1)(\delta - t_2)(\delta - t_3)(\delta - t_4),
\end{aligned}$$

and thence in particular

$$e - d\Sigma\alpha + c\Sigma\alpha\beta - b\Sigma\alpha\beta\gamma + a\alpha\beta\gamma\delta = 0,$$

viz. this is the linear relation which subsists identically between (a, b, c, d, e), the five linear functions of the coordinates (x, y, z, w).

Starting from the above values of (a, b, c, d, e), we find without difficulty

$$\begin{aligned} I = (\alpha-\beta)^2x'y' + (\alpha-\gamma)^2x'z' + (\alpha-\delta)^2x'w' \\ + (\beta-\gamma)^2y'z' + (\beta-\delta)^2y'w' + (\gamma-\delta)^2z'w', \end{aligned}$$

$$\begin{aligned} J = (\alpha-\beta)^2(\alpha-\gamma)^2(\beta-\gamma)^2x'y'z' + (\alpha-\beta)^2(\alpha-\delta)^2(\beta-\delta)^2x'y'w' \\ + (\alpha-\gamma)^2(\gamma-\delta)^2(\alpha-\delta)^2x'z'w' + (\beta-\gamma)^2(\gamma-\delta)^2(\beta-\delta)^2y'z'w', \end{aligned}$$

but we thus see the convenience of introducing constant multipliers into the expressions of the four coordinates respectively, viz. writing

$$X = (\beta\gamma\delta)x', \quad Y = (\gamma\delta\alpha)y', \quad Z = (\delta\alpha\beta)z', \quad W = (\alpha\beta\gamma)w',$$

where for shortness $(\beta\gamma\delta) = (\beta-\gamma)(\gamma-\delta)(\delta-\beta)$, &c., or what is the same thing, taking the quartic to be

$$(a, b, c, d, et, 1)^4 = X'(\beta\gamma\delta)(t+\alpha)^4 + Y'(\gamma\delta\alpha)(t+\beta)^4 + Z'(\delta\alpha\beta)(t+\gamma)^4 + W'(\alpha\beta\gamma)(t+\delta)^4,$$

we find

$$\begin{aligned} J &= (\lambda'\mu'\nu')^2(x'y'z' + y'z'w' + z'x'w' + x'y'w'), \\ I &= (\lambda'\mu'\nu')\{\lambda'(x'w' + y'z') + \mu'(y'w' + z'x') + \nu'(z'w' + x'y')\}, \end{aligned}$$

where for shortness

$$\begin{aligned} \lambda' &= (\alpha-\delta)^2(\beta-\gamma)^2, \\ \mu' &= (\beta-\delta)^2(\gamma-\alpha)^2, \\ \nu' &= (\gamma-\delta)^2(\alpha-\beta)^2, \\ \nabla(\lambda') &= (\alpha-\delta)(\beta-\gamma), \\ \nabla(\mu') &= (\beta-\delta)(\gamma-\alpha), \\ \nabla(\nu') &= (\gamma-\delta)(\alpha-\beta), \\ \nabla(\lambda') + \nabla(\mu') + \nabla(\nu') &= 0, \end{aligned}$$

and the equation of the developable is thus

$$\{\lambda'(x'w' + y'z') + \mu'(y'w' + z'x') + \nu'(z'w' + x'y')\}^3 - 27\lambda'\mu'\nu'(x'y'z' + y'z'w' + z'x'w' + x'y'w')^2 = 0.$$

Observe that $J = 0$ is a cubic surface passing through each edge of the tetrahedron, and having at each summit a conical point; $I = 0$ is a quadric surface passing through each summit of the tetrahedron, and at each of these points the tangent plane of the quadric surface touches the tangent cone of the cubic surface: to show this it is only necessary to observe that at the point $(x' = 0, y' = 0, z' = 0)$ the tangent cone is $y'z' + z'x' + x'y' = 0$, and the tangent plane is $\lambda'x' + \mu'y' + \nu'z' = 0$, and that these touch in virtue of the above-mentioned relation $\nabla(\lambda') + \nabla(\mu') + \nabla(\nu') = 0$.

It follows that on the curve of intersection, or cuspidal edge of the developable, each of the summits is a cuspidal or stationary point, that is, the cuspidal curve has four stationary points; this agrees with the character of the curve as given, Salmon "On the Classification of Curves of Double Curvature," Camb. and Dubl. Math. Jour. vol. V. (1850), p. 39, viz. the character is there given $a = 6$, $m = 6$, $n = 4$, $r = 6$, $g = 3$, $h = 6$, $\alpha = 0$, $\beta = 4$, $\kappa = 4$, $\gamma = 6$, ($\beta = 4$, that is, there are as stated 4 stationary points).

To find the equations of the nodal curve, instead of transforming the equations as given in terms of (a, b, c, d, e), it is better to deduce these from the equation of the surface; viz. if there is a nodal curve, we must have

$$\partial_{x'} I : \partial_{y'} I : \partial_{z'} I : \partial_{w'} I = \partial_{x'} J : \partial_{y'} J : \partial_{z'} J : \partial_{w'} J.$$

Writing these under the form $\partial_{x'} I + \theta' \partial_{x'} J = 0$, &c., where θ' is regarded as an arbitrary parameter[1], we have

$$\begin{aligned}
\lambda' w' + \mu' z' + \nu' y' + \theta'(y'z' + y'w' + z'w') &= 0,\\
\lambda' z' + \mu' w' + \nu' x' + \theta'(z'x' + z'w' + x'w') &= 0,\\
\lambda' y' + \mu' x' + \nu' w' + \theta'(x'y' + x'w' + y'w') &= 0,\\
\lambda' x' + \mu' y' + \nu' z' + \theta'(y'z' + z'x' + x'y') &= 0,
\end{aligned}$$

which equations (eliminating θ') must be equivalent to two equations only.

I remark that the first three equations may be regarded as a set of linear equations in $1, w', \theta', \theta' w'$; and determining from them the ratios of these quantities, we have, suppose,

$$1 : -w' : \theta' : -\theta' w' = A : B : C : D,$$

where

$$A = \begin{vmatrix} \lambda' & \nu' & y' \\ \mu' & \lambda' & z' + x' \\ \nu' & \mu' & x' + y' \end{vmatrix}, \quad B = \begin{vmatrix} y'z' & y' + z' & \nu' z' + \mu' y' \\ z'x' & z' + x' & \lambda' z' + \nu' x' \\ x'y' & x' + y' & \mu' x' + \lambda' y' \end{vmatrix}, \quad \text{etc.}$$

[1]The value of θ' is in fact $= \frac{18J}{I}$, that is, instead of the four equations involving an arbitrary parameter θ', we have really four determinate equations.

We have thence $AD - BC = 0$ and (substituting in the fourth equation) $A(\lambda' x' + \mu' y' + \nu' z') + C(y'z' + z'x' + x'y') = 0$; each of these equations must contain the equation of the cone having $(x' = 0, y' = 0, z' = 0)$ for its vertex, and passing through the nodal curve.

The two equations are of the orders 6 and 4 respectively; and as the curve is a quartic curve passing through the vertex in question, the equation of the cone is of the order 3.

I have not effected the reduction of the sextic equation, but for the quartic equation, substituting for A, C their values, this is

$$\begin{aligned}&- (\lambda' w' + \mu' y' + \nu' z')\{\lambda' x'^2 (y' - z') + \mu' y'^2 (z' - x') + \nu' z'^2 (z' - x')\}\\ &\qquad + (y'z' + z'x' + x'y')\{\lambda' x'(y'^2 - z'^2) + \mu' y'(z'^2 - x'^2) + \nu' z'(x'^2 - y'^2)\}\\ &\qquad\qquad + \{\mu'\nu'(y' - z') + \nu'\lambda'(z' - x') + \lambda'\mu'(x' - y')\}(x' + y' + z') = 0,\end{aligned}$$

which is easily reduced to

$$\begin{aligned}&\lambda' x'(-x'^2 + y'z' + z'x' + x'y')(y' - z')\\ &\qquad + \mu' y'(-y'^2 + y'z' + z'x' + x'y')(z' - x')\\ &\qquad + \nu' z'(-z'^2 + y'z' + z'x' + x'y')(x' - y')\\ &\qquad + \mu'\nu'\{(x' + y' + z')(y'z' + z'x' - x'y') + x'y'z'\}(y' - z')\\ &\qquad + \nu'\lambda'\{(x' + y' + z')(y'z' + z'x' + x'y') + x'y'z'\}(z' - x')\\ &\qquad\qquad + \lambda'\mu'\{(x' + y' + z')(y'z' + z'x' + x'y') + x'y'z'\}(x' - y') = 0;\end{aligned}$$

and I have found that this is transformable into

$$\begin{aligned}&2\{x'^2\nabla(\lambda') + y'^2\nabla(\mu') + z'^2\nabla(\nu')\} \times [x'^2\nabla(\lambda')(\mu' y'^2 - \nu' z'^2)\cdot 2x'\nabla(\mu')(\nu' z'^2 - \lambda' x'^2)\cdot x'y'\nabla(\nu')(\lambda' x'^2 - \mu' y'^2)\\ &\qquad - x'^2 y'^2 z'^2 \{\nabla(\mu') - \nabla(\nu')\}\{\nabla(\nu') - \nabla(\lambda')\}\{\nabla(\lambda') - \nabla(\mu')\}] = 0,\end{aligned}$$

viz. the two functions are equivalent in virtue of the relation $\nabla(\lambda') + \nabla(\mu') + \nabla(\nu') = 0$, or, what is the same thing, they only differ by a function $(x', y', z')^4$ into the evanescent factor $\lambda'^2 + \mu'^2 + \nu'^2 - 2\mu'\nu' - 2\nu'\lambda' - 2\lambda'\mu'$.

The function in $\{\}$ equated to zero is therefore the equation of the cubic cone.

I do not stop to give the steps of the investigation in the above form, as the investigation may be very much simplified as follows: by linear combinations of the four equations in x', y', z', w', θ', we deduce

$$\begin{aligned}&(\lambda' - \mu' - \nu')(x' + w' - y' - z') + 2\theta'(y'z' - x'w') = 0,\\ &(-\lambda' + \mu' - \nu')(y' + w' - z' - x') + 2\theta'(z'x' - y'w') = 0,\\ &(-\lambda' - \mu' + \nu')(z' + w' - x' - y') + 2\theta'(x'y' - z'w') = 0,\\ &(\lambda' + \mu' + \nu')(x' + y' + z' + w') + 2\theta'(y'z' + z'x' + x'y' + x'w' + y'w' + z'w') = 0.\end{aligned}$$

Hence writing

$$\begin{aligned}\lambda &= \lambda' - \mu' - \nu', & x &= w' + x' - y' - z',\\ \mu &= -\lambda' + \mu' - \nu', & y &= w' - x' + y' - z',\\ \nu &= -\lambda' - \mu' + \nu', & z &= w' - x' - y' + z',\\ & & w &= w' + x' + y' + z',\end{aligned}$$

we find

$$\begin{aligned}
\lambda^2 &= \lambda'^2 - \mu'^2 - \nu'^2 + 2\mu'\nu', \\
\mu^2 &= -\lambda'^2 + \mu'^2 - \nu'^2 + 2\nu'\lambda', \\
\nu^2 &= -\lambda'^2 - \mu'^2 + \nu'^2 + 2\lambda'\mu',
\end{aligned}$$

and thence

$$\lambda^2\mu^2 + \mu^2\nu^2 + \nu^2\lambda^2 = -(\lambda'^2 + \mu'^2 + \nu'^2 - 2\mu'\nu' - 2\nu'\lambda' - 2\lambda'\mu')^2 = 0,$$

that is

$$\lambda^2\mu^2 + \mu^2\nu^2 + \nu^2\lambda^2 = 0, \quad \frac{1}{\lambda} + \frac{1}{\mu} + \frac{1}{\nu} = 0,$$

the relation which connects the new constants λ, μ, ν.

Moreover

$$\begin{aligned}
yz - xw &= 4(y'z' - x'w'), \\
zx - yw &= 4(z'x' - y'w'), \\
xy - zw &= 4(x'y' - z'w'), \\
2w^2 - x^2 - y^2 - z^2 &= 8(y'z' + z'x' + x'y' + x'w' + y'w' + z'w'),
\end{aligned}$$

and writing for greater convenience $\theta = -\frac{\theta'}{2}$, the equations are transformed into

$$\begin{aligned}
\theta\lambda x &= vw - yz, \\
\theta\mu y &= yw - zx, \\
\theta\nu z &= zw - xy, \\
2\theta(\lambda + \mu + \nu)w &= 3w^2 - x^2 - y^2 - z^2,
\end{aligned}$$

where

$$\frac{1}{\lambda} + \frac{1}{\mu} + \frac{1}{\nu} = 0,$$

viz. these equations, eliminating θ, give the equations of the nodal curve.

From the first three equations eliminating θ, we deduce

$$\begin{aligned}
yzw(\mu - \nu) &= x(\mu y^2 - \nu z^2), \\
zxw(\nu - \lambda) &= y(\nu z^2 - \lambda x^2), \\
xyw(\lambda - \mu) &= z(\lambda x^2 - \mu y^2),
\end{aligned}$$

or, as these equations may be written,

$$\frac{x(\mu y^2 - \nu z^2)}{\mu - \nu} = \frac{y(\nu z^2 - \lambda x^2)}{\nu - \lambda} = \frac{z(\lambda x^2 - \mu y^2)}{\lambda - \mu},$$

which equations, from the mode in which they are obtained, are it is clear equivalent to two equations only.

Using the fourth equation, and eliminating by substituting therein for $\theta\lambda$, $\theta\mu$, $\theta\nu$ their values from the first three equations, we find

$$2w^2\left(3w^2 - x^2 - y^2 - z^2\right) = \frac{3w^2 - x^2 - y^2 - z^2}{\lambda} + \frac{3w^2 - x^2 - y^2 - z^2}{\mu} + \frac{3w^2 - x^2 - y^2 - z^2}{\nu}$$

that is

$$3w^2 + x^2 + y^2 + z^2 = 2w\left(\frac{x^2}{\lambda} + \frac{y^2}{\mu} + \frac{z^2}{\nu}\right),$$

or, what is the same thing,

$$xyz(3w^2 - x^2 + y^2 + z^2) - 2w(y^2z^2 + z^2x^2 + x^2y^2) = 0,$$

we have to show that this is in fact included in the former system, for then the four equations with θ eliminated will it is clear give two equations only.

Observe that the former system may be written

$$\frac{x(\mu y^2 - \nu z^2)}{(\mu - \nu)yz} = \frac{y(\nu z^2 - \lambda x^2)}{(\nu - \lambda)zx} = \frac{z(\lambda x^2 - \mu y^2)}{(\lambda - \mu)xy},$$

$$\begin{aligned}(\mu - \nu)yz \cdot x(\mu y^2 - \nu z^2) + (\nu - \lambda)zx \cdot y(\nu z^2 - \lambda x^2) + (\lambda - \mu)xy \cdot z(\lambda x^2 - \mu y^2)\\
= \mu\nu yz(\mu - \nu)(y^2 - z^2) + \nu\lambda zx(\nu - \lambda)(z^2 - x^2) + \lambda\mu xy(\lambda - \mu)(x^2 - y^2)\\
+ xyz\{\lambda(\mu - \nu)(\mu y^2 + \nu z^2) + \mu(\nu - \lambda)(\nu z^2 + \lambda x^2) + \nu(\lambda - \mu)(\lambda x^2 + \mu y^2)\}.\end{aligned}$$

[The paper continues on pp. 373—376 with further development of the theory of the sextic developable and its connection with the sextic surface. The investigation involves detailed analysis of the nodal curve, the cuspidal edge, and their singularities, together with the coordinate transformations and algebraic relations connecting the different forms of the surface equations.]

399. On the Cubical Divergent Parabolas

[From the Quarterly Journal of Pure and Applied Mathematics, vol. ix. (1868), pp. 185—190.]

[From p. 101.]

[Pp. 185—190.]

The cubical divergent parabolas are the parabolas $y^2 = x^3 + ax + b$, where the parameters a, b are such that the discriminant $4a^3 + 27b^2$ is $= 0$; viz. these are curves for which the line at infinity meets the curve in three distinct points, but the line at infinity is a tangent, or the curve has a cusp at infinity.

The several forms may be distinguished according to the nature of the roots of the equation $4a^3 + 27b^2 = 0$ considered as a cubic equation in $a : b$; or what is the same thing, according to the nature of the roots of the equation $z^3 - 3z + 2 = 0$ (where for convenience a, b have been replaced by $3z$, -2 respectively), viz. the roots are $z = 1, 1, -2$; and the forms correspond to the two cases $z = 1$ and $z = -2$.

For $z = 1$, that is $4a^3 + 27b^2 = 0$, $a \neq 0$, the equation is $y^2 = x^3 - 3x + 2$, that is $y^2 = (x-1)^2(x+2)$, which has a crunode at infinity; or writing $x = \frac{1}{\xi}$, $y = \frac{\eta}{\xi^2}$, the equation becomes $\eta^2 = (1-\xi)^2(1+2\xi)$, which has a crunode at $\xi = 0$, $\eta = 0$, viz. the origin.

For $z = -2$, that is $a = 0$, $b = 0$, the equation is $y^2 = x^3$, which has a cusp at infinity; or in the like transformed equation $\eta^2 = \xi^3$, which has a cusp at the origin.

The form $y^2 = x^3 - 3x + 2$, that is $y^2 = (x-1)^2(x+2)$, has a crunode at infinity; the tangent at the crunode is the line at infinity, and the two branches are each touched by the line at infinity. The form $y^2 = x^3$ has a cusp at infinity, the tangent at the cusp being the line at infinity.

It is convenient to exhibit the different species in a tabular form, indicating the nature of the singularities (finite and at infinity), and the sign of the discriminant $\Delta = -(4a^3 + 27b^2)$ in the different cases.

Equation	**Singularities**	Δ
$y^2 = x^3 + 3x - 2$	simplex acnodal, R without the curve	$-$
$y^2 = x^3$	simplex cuspidal, R at the cusp	0
$y^2 = x^3 - 3x + 2$	simplex crunodal, R at the node	$+$
$y^2 = x^3 + ax + b$ $(4a^3 + 27b^2 \neq 0)$	simplex non-singular	$\neq 0$

The species may be further subdivided according to the position of the point R (the intersection of the three inflexions) relative to the curve and its singularities. For the simplex non-singular form, we have according as R is within or without the curve, the campaniform or serpentine forms; and for the singular forms, corresponding subdivisions according as R coincides with or is distinct from the singularity, and in the latter case according as it lies within or without the curve.

400. On the Cubic Curves Inscribed in a Given Pencil of Six Lines

[From the Quarterly Journal of Pure and Applied Mathematics, vol. ix. (1868), pp. 211—230.]

[From p. 105.]

[Pp. 211—230.]

I consider a system of six lines forming a pencil, that is, meeting in a point, and I seek to investigate the general equation of a cubic curve inscribed in the six lines; viz. the six lines are to be tangents to the cubic curve. It is to be observed that the six lines may be any six lines of a pencil; there is no restriction whatever as to the particular positions of the lines. The problem is: given six tangents of a cubic, to find the equation of the cubic; or what is the same thing, to find the equation of the curve having the six given lines for tangents.

Taking the centre of the pencil for the point $(x = 0, y = 0)$, and taking $z = 0$ for the equation of the line at infinity, the equations of the six tangents may be written

$$z = 0, \quad y - m_1 x = 0, \quad y - m_2 x = 0, \quad \ldots, \quad y - m_5 x = 0,$$

where m_1, m_2, m_3, m_4, m_5 are the slopes of the five finite tangents respectively. The general equation of a cubic curve is

$$(a, b, c, d, f, g, h, l, m, nx, y, z)^3 = 0,$$

or what is the same thing,

$$ax^3 + by^3 + cz^3 + 3dy^2z + 3ez^2x + 3fz^2y + 3gxz^2 + 3hyz^2 + 3lx^2y + 3mxy^2 + 6nxyz = 0.$$

The condition that the line $y - mx = 0$ may be a tangent to this cubic is obtained by substituting $y = mx$ in the equation of the cubic, and expressing that the resulting equation in $x : z$ has equal roots; viz. substituting $y = mx$, we have

$$(a + 3lm + 3m^2 + bm^3)x^3 + 3(n + fm + em^2 + hm^3 + gm^2 + dm^3)xz^2 + (c + 3gz + 3hz + dz)z^3 = 0,$$

or more simply

$$(a, b, c, d, e, f, g, h, l, m, n1, m, 0)^3 x^3 + \cdots = 0.$$

The condition of equal roots gives a sextic equation in m, which must be satisfied identically for $m = m_1, m_2, m_3, m_4, m_5$ and $m = \infty$ (corresponding to the tangent $z = 0$). We hence obtain the required conditions on the coefficients.

The investigation is conducted by means of the invariants of the cubic. Writing the cubic in the canonical form, and making use of the known relations between the invariants and the coefficients, the conditions that six given lines of a pencil shall be tangents to the cubic may be expressed in a very elegant form.

Let the six tangents be denoted by their equations $l_1 = 0, l_2 = 0, \ldots, l_6 = 0$, where $l_i = y - m_i x$ for $i = 1, \ldots, 5$, and $l_6 = z$. Then the equation of any cubic having these six lines for tangents may be written in the form

$$\sqrt{l_1} + \sqrt{l_2} + \sqrt{l_3} + \sqrt{l_4} + \sqrt{l_5} + \sqrt{l_6} = 0,$$

where the signs of the radicals are properly determined.

Or more definitely, if we write

$$\sqrt{l_1} \pm \sqrt{l_2} \pm \sqrt{l_3} \pm \sqrt{l_4} \pm \sqrt{l_5} \pm \sqrt{l_6} = 0,$$

the different systems of signs give the different cubics which have the six lines for tangents. It is known that there are in general ten such cubics; the different systems of signs must be such that the product of all the terms is positive, and the number of independent systems is ten.

The equation may also be written in the rationalised form. We have

$$(\sqrt{l_1} + \sqrt{l_2} + \sqrt{l_3} + \sqrt{l_4} + \sqrt{l_5} + \sqrt{l_6})(\sqrt{l_1} + \sqrt{l_2} + \sqrt{l_3} + \sqrt{l_4} + \sqrt{l_5} - \sqrt{l_6}) \cdots = 0,$$

where the product contains $2^5 = 32$ factors, but these combine in pairs to give 16 different cubics, and of these only 10 are proper cubics, the remaining 6 being degenerate forms.

For the complete discussion it is necessary to introduce the fundamental invariants S and T of the cubic. These are connected with the coefficients by the relations

$$S = \frac{1}{3 \cdot 12^4}(abc + \dots),$$
$$T = \frac{1}{27 \cdot 12^6}(a^2b^2c^2 + \dots),$$

and the discriminant is $\Delta = 64S^3 - T^2$.

For a cubic having six given tangents which form a pencil, the invariants S and T satisfy certain relations which may be obtained as follows. Let the six lines be given, and let the cubic be expressed in terms of the parameters which determine the position of the point of contact on each tangent; then the condition that the six points of contact shall lie on a cubic curve gives the required relation between S and T.

The detailed calculation is as follows. Let the cubic be written in the form

$$c[(a, h, k, bx, y)^2]^2 + 32l\Delta x^2y^2 = 0,$$

where $(a, h, k, bx, y)^2 = ax^2 + 2hxy + by^2$, and $\Delta = hk - ab$, $V = hk - h^2$, $\Theta = ah - bk$, $R = -81ab$.

The discriminant of this form, say Q_{10}, is a function of c^2D, V, $c\Theta$, cR, where D is the discriminant of the quadratic $(a, h, k, bx, y)^2$. The expression is found to be

$$\begin{aligned} Q_{10} = c^{10}\{D^3\} + c^9\{-99RD^2 - 400VD + 2304V\Theta D + 8640\Theta^2\} \\ + c^8\{12800R^2V\Theta + 82944V^3\Theta^2 - 4608V^2D\} \\ + c^7\{20480R^2V^3 + 147456V^4\Theta + 65536V^6\}. \end{aligned}$$

The foregoing values of S and T give

$$T^2 - 64S^3 = c^4\{D^3 + c(8RD^2 - 24V\Theta D - 64\Theta^3) + c^2(16R^2 + 96RV\Theta - 48V^2\Theta^2 + 16V^3D) + c^3(64R^2V^2)\},$$

so that

$$Q_{10} = -c^6R^2(T^2 - 64S^3),$$

and therefore

$$64S^3 - T^2 = \frac{Q_{10}}{c^6R^2},$$

formulae which are interesting in the theory.

As regards the discriminant Q_{10}, this as already remarked, has been calculated for the general form, but for the present purpose it is easier, by dealing directly with the form $[(a, h, k, bx, y)^2]^2 + 32l\Delta x^2y^2$ and then interpreting D, V, Θ, R and restoring the coefficient c as above, to obtain the discriminant Q_{10} of the function $[c(a, h, k, bx, y)^2]^2 + 4[(j, l, fx, y)^2]^2$ in the required form, as a function of c^2D, V, $c\Theta$, cR.

I find after some laborious calculations

$$\begin{aligned}
Q_5 &= \text{No. } 31 = c^5\{90\} + c^4\{40VD - 288V\Theta\} + c^3\{256V^2\},\\
Q_7 &= 10000\,\text{No. } 34 = c^7\{-99D^2 - 400RD + 2304V\Theta D + 8640\Theta^2\}\\
&\quad + c^6\{12800R^2V\Theta + 82944V^3\Theta^2 - 4608V^2D\} + c^5\{20480R^2V^3 + 147456V^4\Theta + 65536V^6\},\\
Q_8 &= 1000000\,\text{No. } 35 = c^8\{7992D^3 + 72000RD^2 - 145152V\Theta D^2 + 622080\Theta^2D\}\\
&\quad + c^7\{160000R^2\Theta + 691200R^2V\Theta D + 3456000R^2\Theta^2 + 3815424V^2\Theta^2D\\
&\qquad + 36080640V\Theta^3 + 635904V^2D^2\}\\
&\quad + c^6\{33177600R^2V^2\Theta^2 + 4669440R^2V^2D + 217645056V^4\Theta^2 + 23003136V^2\Theta D\}\\
&\quad + c^5\{62914560R^2V^4 + 452984832V^3\Theta + 14155776V^4D\}\\
&\quad + c^4\{134217728V^8\}.
\end{aligned}$$

Also Q_{10} = multiple of discriminant $= c^{10}R^2U^2$, where

$$\begin{aligned}
U = c^4D^3 &+ c^3(-8R^2D - 24R^2V\Theta D + 64R^2\Theta^3)\\
&+ c^2(-16R^2 - 96R^2V\Theta + 48R^2V^2\Theta^2 - 16R^2V^3D) + c(-64R^2V^2).
\end{aligned}$$

To these may be joined the intermediate values:

$$\begin{aligned}
Q_3 &= c^3\{81D\} + c^2\{-720RD + 5184V\Theta D + 1600R^2 + 23040RV\Theta + 82944V^2\Theta^2 + 4608V^2D\}\\
&\quad + c\{20480R^2V^3 + 147456V^4\Theta + 65536V^6\},\\
Q_4 &= c^4\{180D^2 + 1120RD + 2880V\Theta D - 8640\Theta^2\}\\
&\quad + c^3\{1600R^2 + 10240RV\Theta + 270D^2 + 720RD^2 - 51840V\Theta D^2 - 77760\Theta^2D\}.
\end{aligned}$$

We have

$$c(fO) = T - 12VS + 4V^3 - 4cR,$$

and if by means of these values we eliminate $c\Theta$ and c^2D, we obtain Q_5, Q_7, Q_8 and Q_{10} as functions of S, T, V and cR. Choosing instead of Q_5 and Q_8 the combinations $Q_5 - Q_3$ and $Q_8 - 8Q_4^2$, and forming also the expression for the combination $Q_5(Q_5 - Q_3)$, we have thus the system of formulae:

$$\begin{aligned}
Q_3 &= 9T^2 + 180VS + 4V^3 + 4cR,\\
\tfrac{1}{4}(Q_5 - Q_3) &= 9T^2 - 432S^3 - 72TVS - 72TV - 16TcR + 864V^2S^2 + 144V^2S + 128VScR,\\
\tfrac{1}{4}(Q_8 - 8Q_4^2) &= 27T^4 - 4212T^2VS^2 + 6588T^2V^2 + 252T^2cR - 7776TS^3\\
&\quad - 16848TV^2S^2 + 3456TV^2S - 2592TV^2ScR - 1296TV - 1824TV^2cR\\
&\quad - 448Tc^2R^2 + 544320VS^4 - 461376V^2S^3 + 74304cRS^3 + 15552V^2S^2\\
&\quad - 10368V^2S^2cR - 10728V^2S - 3264V^2ScR - 1536V^2c^2R^2,\\
&\quad + 81T^4 + 972T^3VS - 612T^3V - 108T^3cR - 3888TS^3 - 5184TV^2S^2\\
&\quad - 11952TV^2S - 2016TVScR - 288TV - 352TV^2cR - 64Tc^2R\\
&\quad - 77760VS^4 + 153792V^2S^3 - 1728cRS^3 + 29376V^2S^2 + 26928V^2S^2cR\\
&\quad + 576V^2S + 1088V^2ScR + 512V^2c^2R^2.
\end{aligned}$$

And, as mentioned above, $Q_{10} = -c^6R^2(T^2 - 64S^3)$.

The just-mentioned value of Q_{10} should, I think, admit of being established *a priori*, and if this be so, then the substitution of the values of S and T in terms of c^2D, V, $c\Theta$, cR, would be the easiest way of arriving at the before-mentioned expression of Q_{10} in terms of these same quantities. The calculation by which this expression was arrived at, is however not without interest, and it will be as well to indicate the mode in which it was effected.

Calculation of Q_{10}.

We have to find the discriminant of $c[(a,h,k,bx,y)^2]^2 + 32l\Delta x^2y^2$.

Consider for a moment the more general form P^2+4Q^2, then to find the discriminant, we have to eliminate between the equations

$$P\frac{\partial P}{\partial x} + 4Q\frac{\partial Q}{\partial x} = 0, \quad P\frac{\partial P}{\partial y} + 4Q\frac{\partial Q}{\partial y} = 0;$$

these are satisfied by the system $P=0$, $Q^2=0$, and it follows that if R be the resultant of the equations $P=0$, $Q=0$, then the discriminant in question contains the factor R^2. For the other factor we may reduce the system to

$$P\frac{\partial P}{\partial x} + 4Q\frac{\partial Q}{\partial x} = 0, \quad \frac{\partial P}{\partial x}\frac{\partial Q}{\partial y} - \frac{\partial P}{\partial y}\frac{\partial Q}{\partial x} = 0.$$

Now writing $Q = 2lxy$, these equations become

$$P\frac{\partial P}{\partial x} + 4\cdot 8l^2xy^2 = 0, \quad \left(\frac{\partial P}{\partial x}\frac{\partial P}{\partial y}\right) = 0,$$

the resultant of which is $=$ resultant of the system

$$P^2 + 4\cdot 8l^2x^2y^2 = 0, \quad \frac{\partial P}{\partial x}\frac{\partial P}{\partial y} = 0,$$

but in virtue of the second equation, we have

$$\frac{\partial P}{\partial x}\frac{\partial P}{\partial y} = 0,$$

which reduces the first equation to

$$P^2 + 32l^2x^2y^2 = 0,$$

or omitting the factor $2y$, to

$$\frac{\partial P}{\partial x} = 0.$$

Hence, writing $P = \sqrt{c}\cdot(a, h, k, bx, y)^2$, and therefore $\frac{\partial P}{\partial x} = 3\sqrt{c}\cdot(a, h, k, bx, y)^2$, $\frac{\partial P}{\partial y} = 3\sqrt{c}\cdot(h, k, bx, y)^2$; writing also $y = 1$, the two equations become

$$\begin{aligned} c(a, h, k, bx, 1)^2(h, k, bx, 1)^2 + 8l^2x^2 &= 0,\\ (a, h, k, bx, 1)^2 - (h, k, bx, 1)^2 &= 0, \end{aligned}$$

the second of which is more simply written

$$(a, h, -k, -bx, 1)^2 = 0.$$

Hence, restoring the factor P, and also to avoid fractions introducing the factor $8a^2$, the resultant of the two equations is

$$= 8l^2a^2\prod(8l^2x^2 + c(a, h, k, bx, 1)^2(h, k, bx, 1)^2),$$

where $\prod$ denotes the product of the factors corresponding to the three roots x_1, x_2, x_3 of the equation

$$(a, h, -k, -bx, 1)^2 = 0, \quad \text{or} \quad ax^3 + hx^2 - kx - b = 0,$$

so that the symmetric functions are to be found from

$$\Sigma x_1 = -\frac{h}{a}, \quad \Sigma x_1x_2 = -\frac{k}{a}, \quad x_1x_2x_3 = \frac{b}{a}.$$

The required discriminant is the foregoing resultant multiplied by R, or say by c^2E^2, that is the discriminant Q_{10} is

$$= c^2E^2 \cdot 8l^2a^2 \prod(8l^2x^2 + \Omega),$$

if for shortness we write

$$\Omega = (a, h, k, bx, 1)^2 \cdot (a, k, bx, 1)^2,$$

and when the symmetric functions have been expressed in terms of the coefficients, the result is to be expressed as a function of D, V, Θ, R by means of the values

$$\begin{aligned} c^2D &= a^2b^2 + 4ahk^2 + 4bk^2h - 6abhk - 3h^2k^2, \\ c\Theta &= -l(ab - hk), \\ cR &= -81l^2ab. \end{aligned}$$

Thus, for instance, the first term of the result is

$$= c^2R^2 \cdot 8l^2a^2 \cdot 512l^6x_1^2x_2^2x_3^2,$$

which is

$$= c^2R^2 \cdot 4096l^8 \cdot \frac{b^2}{a^2} = -64c^4R^4V^2,$$

which is a term in the before-mentioned expression for Q_{10}.

401. A Notation of the Points and Lines in Pascal's Theorem

[From the Quarterly Journal of Pure and Applied Mathematics, vol. ix. (1868), pp. 268—274.]

Taking six points 1, 2, 3, 4, 5, 6 on a conic; let $A, B, C, D, E, F, G, H, I, J, K, L, M, N, O$ denote each a combination of three lines, thus

$12 \cdot 34 \cdot 56 = A$	$13 \cdot 45 \cdot 62 = B$	$14 \cdot 56 \cdot 23 = C$
$15 \cdot 62 \cdot 34 = D$	$16 \cdot 23 \cdot 45 = E$	$12 \cdot 35 \cdot 64 = F$
$13 \cdot 46 \cdot 25 = G$	$14 \cdot 52 \cdot 36 = H$	$15 \cdot 63 \cdot 42 = I$
$16 \cdot 24 \cdot 53 = J$	$12 \cdot 36 \cdot 45 = K$	$13 \cdot 42 \cdot 56 = L$
$14 \cdot 53 \cdot 62 = M$	$15 \cdot 64 \cdot 23 = N$	$16 \cdot 25 \cdot 34 = O$

then any hexagon formed with the six points may be represented by a combination of some two of the letters A, B, &c., viz. the three alternate sides are the lines represented by one letter, and the other three alternate sides the lines represented by the other letter: for example, the hexagon 123456 is AE; and so for the other hexagons.

Any duad AE thus representing a hexagon may be termed a *hexagonal duad*; the number of such duads is sixty.

Each Pascalian line may be denoted by the symbol of the hexagon to which it belongs; thus, the line which belongs to the hexagon AE, is the line AE.

I form the following combinations:

$IMO \cdot DHJ$	each involving all the duads 12, &c. except those of $123 \cdot 456$,
$DEG \cdot BNO$	$124 \cdot 356$,
$ELM \cdot BGJ$	$125 \cdot 346$,
$HLN \cdot CGI$	$126 \cdot 345$,
$EFI \cdot JKN$	$134 \cdot 256$,
$AEH \cdot GKO$	$135 \cdot 246$,
$AMN \cdot GDF$	$136 \cdot 245$,
$AGJ \cdot ELO$	$145 \cdot 236$,
$ABI \cdot DKL$	$146 \cdot 235$,
$OKM \cdot BFH$	$156 \cdot 234$,

and also the combinations:

$AEGMI$	involving all the duads 12, 13, &c.,
$ABHJN$	
$BGFIO$	
$CDGJK$	
$DEFHL$	
$KLMNO$	

which I call respectively the *ten-partite* and *six-partite* arrangements.

It is to be remarked that (considering $IMO \cdot DHJ$ as standing for the six duads IM, IO, MO, DH, DJ, HJ, and so for the others) the ten-partite arrangement contains all the sixty hexagonal duads: and in like manner, (considering $AEGMI$ as standing for the ten duads AE, AG, AM, AI, EG, EM, EI, GM, GI, MI, and so for the others) the six-partite arrangement contains all the sixty hexagonal duads.

The 60 Pascalian lines intersect by 4's in the 45 Pascalian points p, by 3's in 20 points g and in 60 points h, and by 2's in 90 points m, 360 points r, 360 points t, 360 points z, and 90 points w.

The intersections of the Pascalian lines thus are

$45p$	counting as	270
$20g$	"	40
$60h$	"	180
$90m$	"	90
$360r$	"	360
$360t$	"	360
$360z$	"	360
$90w$	"	90
$1770 = 60 \cdot 59,$		

and the intersections on each Pascalian line are

$9p$	counting as	9
$2g$	"	2
$3h$	"	6
$3m$	"	3
$12r$	"	12
$12t$	"	12
$12z$	"	12
$3w$	"	3
59.		

For the ten-partite arrangement, any double triad such as $ABI \cdot DKL$ gives 15 intersections; $10 \times 15 = 150$; and any pair of double triads such as $ABI \cdot DKL$ and $AEH \cdot GKO$ gives 36 intersections; $45 \times 36 = 1620$; and these are

$$\begin{array}{c}
10 \times 6g = 60g \ 10 \times 15p = 150 \\
45 \times 2h = 90h \ 45 \times 36 = 1620 \\
180h \\
8r = 360r \\
8t = 360t \\
8z = 360z \\
2m = 90w \\
\hline
150 \ 36 \\
1620 \\
1770.
\end{array}$$

For the six-partite arrangement any pentad such as $ABHJN$ gives 45 intersections; $6 \times 45 = 270$; and any two pentads such as $ABHJN$ and $AEGMI$ give 100 intersections; $15 \times 100 = 1500$; and these are

$$\begin{array}{c}
6 \times 15p = 90 \ 15 \times 6h = 90h \\
60g \ 15 \times 24r = 360r \\
180h \ 15 \times 24t = 360t \\
90m \ 15 \times 24z = 360z \\
45 \times 2 = 90w \\
\hline
15 \times 18p = 270 \ 100 \\
1500 \\
1770.
\end{array}$$

I analyse the intersections of a Pascalian line, say AE, by the remaining 59 Pascalian lines as follows: Observe that AE belongs to the triad AEH, the complementary triad whereof is GKO; it also belongs to the pentad $AEIMG$. We thus obtain, corresponding to AE, the arrangement

$$\begin{array}{cccccc}
 & H & H & H & H & A \\
 & B & N & J & & \\
\text{cline1-3} & E & F & L & D & \\
\text{cline1-4} & I & M & G & & K \\
 & & & & C & O
\end{array}$$

viz. $HABNJ$, is the pentad which contains HA, the arrangement of the last three letters B, N, J thereof being arbitrary; $HEFLD$ is the pentad that contains HE, but the last three letters are so arranged that the columns HBF, HNL, HJD are each of them a triad, IMG is then the residue of the pentad $AEIMG$, and KGO is the complementary triad to AEH, but the arrangement of the letters IMG, and of the letters KGO, are each of them determinate; viz. these are such that we have $BFIGO$, $NLMKO$, $JDGCK$, each of them a pentad.

And this being so we derive from the arrangement

$2g$	AH, EH;
$3m$	KG, KC, GO;
$6h$	AI, AM, AG; EI, EM, EG;
$12z$	IB, IF, MN, ML, GJ, GD;
	HB, HF, HN, HL, HJ, HD;
$9p$	AB, AN, AJ; EF, EL, ED; BF, NL, JD;
$12r$	CB, OF, GJ, GD; OB, OF, ON, OL;
	KN, KL, KJ, KD;
	FL, FD, LD; BN, BJ, NJ;
$12t$	IG, IO; MK, MO; GK, GC;
$3m$	IM, IC, MG;
$3w$	CD, EK, LM;

viz. the line AE in question meets AH, EH each of them in a point g; KG, KC, GO each in a point m; and so on.

By constructing in the same way an arrangement for each of the lines AH, &c., we find the nature of the point of intersection of any two of the lines AB, AE, AH, &c.; and we may then present the results in a table (see Plate), which shows at a glance what is the point of intersection (whether a point g, m, h, z, p, r, t, or w) of any two of the Pascalian lines.

I further remark that representing the 45 Pascalian points as follows:

$12 \cdot 34 = a$	$12 \cdot 35 = b$	$12 \cdot 36 = c$	$12 \cdot 45 = d$	$12 \cdot 46 = e$	$12 \cdot 56 = f$
$23 \cdot 45 = c$	$23 \cdot 46 = g$	$23 \cdot 56 = h$	$24 \cdot 35 = j$	$24 \cdot 36 = k$	$24 \cdot 56 = l$
$13 \cdot 24 = g$	$13 \cdot 25 = h$	$13 \cdot 26 = i$	$13 \cdot 45 = j$	$13 \cdot 46 = k$	$13 \cdot 56 = l$
$25 \cdot 34 = l$	$25 \cdot 36 = m$	$25 \cdot 46 = n$	$26 \cdot 34 = p$	$26 \cdot 35 = q$	$26 \cdot 45 = r$
$14 \cdot 23 = m$	$14 \cdot 25 = n$	$14 \cdot 26 = o$	$14 \cdot 35 = p$	$14 \cdot 36 = q$	$14 \cdot 56 = r$
$34 \cdot 56 = p$	$35 \cdot 46 = s$	$36 \cdot 45 = t$			
$15 \cdot 23 = s$	$15 \cdot 24 = t$	$15 \cdot 26 = u$	$15 \cdot 34 = v$	$15 \cdot 36 = w$	$15 \cdot 46 = x$
$16 \cdot 23 = y$	$16 \cdot 24 = z$	$16 \cdot 25 = \alpha$	$16 \cdot 34 = \beta$	$16 \cdot 35 = \gamma$	$16 \cdot 45 = \delta$

the sixty hexagons and their Pascalian lines then are

AE	123456	$12 \cdot 45\ 23 \cdot 56\ 34 \cdot 61$	$d\nu\pi$
AH	125634	$12 \cdot 63\ 25 \cdot 34\ 56 \cdot 41$	$c\lambda r$
EH	145236	$14 \cdot 23\ 45 \cdot 36\ 52 \cdot 61$	mqa
CK	123654	$12 \cdot 65\ 23 \cdot 54\ 36 \cdot 41$	M
CO	143256	$14 \cdot 25\ 43 \cdot 56\ 32 \cdot 61$	npy
KO	125436	$12 \cdot 43\ 25 \cdot 36\ 54 \cdot 61$	$a\mu h$
AM	126534	$12 \cdot 53\ 26 \cdot 34\ 65 \cdot 41$	$b\lambda r$
AG	125643	$12 \cdot 64\ 25 \cdot 43\ 56 \cdot 31$	$e\backslash l$
AI	124365	$12 \cdot 36\ 24 \cdot 65\ 43 \cdot 51$	$c\kappa v$
EG	132546	$13 \cdot 54\ 32 \cdot 46\ 25 \cdot 61$	$j?a$
BF	126435	$12 \cdot 43\ 26 \cdot 35\ 64 \cdot 51$	amx
FL	124653	$12 \cdot 65\ 24 \cdot 53\ 46 \cdot 31$	fOk
DL	134265	$13 \cdot 26\ 34 \cdot 65\ 42 \cdot 51$	$i\rho t$
BN	132645	$13 \cdot 64\ 32 \cdot 45\ 26 \cdot 51$	$k\epsilon u$
BJ	135426	$13 \cdot 42\ 35 \cdot 26\ 54 \cdot 61$	$g\omega\xi$
JN	153246	$15 \cdot 24\ 53 \cdot 46\ 32 \cdot 61$	$t\alpha\gamma$
GK	125463	$12 \cdot 46\ 25 \cdot 63\ 54 \cdot 31$	$e/\eta j$
KM	126354	$12 \cdot 35\ 26 \cdot 54\ 63 \cdot 41$	$h\iota rq$
IO	152436	$15 \cdot 43\ 52 \cdot 36\ 24 \cdot 61$	$\nu\epsilon\alpha$
MO	143526	$14 \cdot 52\ 43 \cdot 26\ 35 \cdot 61$	$n\zeta h$
EM	145326	$14 \cdot 32\ 45 \cdot 26\ 53 \cdot 61$	$r\iota\pi'\zeta$
EI	154236	$15 \cdot 23\ 54 \cdot 36\ 42 \cdot 61$	$s\tau\xi$
AN	123465	$12 \cdot 46\ 23 \cdot 65\ 34 \cdot 51$	$e\eta\nu$
AJ	124356	$12 \cdot 35\ 24 \cdot 56\ 43 \cdot 61$	$b\kappa\lambda$
AB	126543	$12 \cdot 54\ 26 \cdot 43\ 65 \cdot 31$	$d\zeta l$
DE	154326	$15 \cdot 32\ 54 \cdot 26\ 43 \cdot 61$	$s\tau\tau'\zeta$
EL	132456	$13 \cdot 45\ 32 \cdot 56\ 24 \cdot 61$	$j\eta\xi$
EF	123546	$12 \cdot 54\ 23 \cdot 46\ 35 \cdot 61$	$d\phi y$
CD	143265	$14 \cdot 26\ 43 \cdot 65\ 32 \cdot 51$	$op\xi$
CF	123564	$12 \cdot 56\ 23 \cdot 64\ 35 \cdot 41$	$A\rho$

CG	132564	$13 \cdot 56\ 32 \cdot 64\ 25 \cdot 41$	$l\mu n$
CI	142365	$14 \cdot 36\ 42 \cdot 65\ 23 \cdot 51$	$q\iota c\xi$
MN	146235	$14 \cdot 23\ 46 \cdot 35\ 62 \cdot 51$	$ma\alpha$
GJ	135246	$13 \cdot 24\ 35 \cdot 46\ 52 \cdot 61$	$ga\alpha L$
BI	136245	$13 \cdot 24\ 36 \cdot 45\ 62 \cdot 51$	gru
DG	134625	$13 \cdot 62\ 34 \cdot 25\ 46 \cdot 51$	$i\kappa\alpha\xi$
LM	135624	$13 \cdot 62\ 35 \cdot 24\ 56 \cdot 41$	$i\lambda r$
FI	124635	$12 \cdot 63\ 24 \cdot 35\ 46 \cdot 51$	cdx
BH	136254	$13 \cdot 25\ 36 \cdot 54\ 62 \cdot 41$	$hr\alpha$
FH	125364	$12 \cdot 36\ 25 \cdot 64\ 53 \cdot 41$	cvp
FO	125346	$12 \cdot 34\ 25 \cdot 46\ 53 \cdot 61$	avy
LO	134256	$13 \cdot 25\ 34 \cdot 56\ 42 \cdot 61$	hpz
DK	126345	$12 \cdot 34\ 26 \cdot 45\ 63 \cdot 51$	$a\iota rw$
KL	124563	$12 \cdot 56\ 24 \cdot 63\ 45 \cdot 31$	$f\xi j$
BO	134526	$13 \cdot 52\ 34 \cdot 26\ 45 \cdot 61$	$h\beta\xi$
NO	152346	$15 \cdot 34\ 52 \cdot 46\ 23 \cdot 61$	νvy
BC	132654	$13 \cdot 65\ 32 \cdot 54\ 26 \cdot 41$	leo
CJ	142356	$14 \cdot 35\ 42 \cdot 56\ 23 \cdot 61$	$p\kappa y$
JK	124536	$12 \cdot 53\ 24 \cdot 36\ 45 \cdot 61$	$h\delta$
KN	123645	$12 \cdot 64\ 23 \cdot 45\ 36 \cdot 51$	$e\epsilon w$
DH	143625	$14 \cdot 62\ 43 \cdot 25\ 36 \cdot 51$	$o\lambda w$
HJ	142536	$14 \cdot 53\ 42 \cdot 36\ 25 \cdot 61$	$p\iota a$
HL	136524	$13 \cdot 52\ 36 \cdot 24\ 65 \cdot 41$	$h\iota r$
HN	146325	$14 \cdot 32\ 46 \cdot 25\ 63 \cdot 51$	$m\nu w$
BF	126453	$12 \cdot 46\ 26 \cdot 53\ 64 \cdot 31$	$dc\alpha k$
DJ	153426	$15 \cdot 42\ 53 \cdot 26\ 34 \cdot 61$	$t\alpha\xi$
LN	132465	$13 \cdot 46\ 32 \cdot 65\ 24 \cdot 51$	$k\eta t$
GM	135264	$13 \cdot 26\ 35 \cdot 64\ 52 \cdot 41$	$i\iota rn$
IM	142635	$14 \cdot 63\ 42 \cdot 35\ 26 \cdot 51$	qdv
GI	136425	$13 \cdot 42\ 36 \cdot 25\ 64 \cdot 51$	$g\mu M$

Each Pascalian point belongs to four different hexagons; viz.

a to the hexagons	$\{K,F\}\{D,O\}$	n	$\{B,L\}\{F,N\}$
b	$\{A,K\}\{M,J\}$	o	$\{A,C\}\{B,G\}$
c	$\{A,F\}\{H,I\}$	p	$\{B,D\}\{G,H\}$
d	$\{A,F\}\{B,E\}$	q	$\{C,M\}\{I,K\}$
e	$\{A,K\}\{G,N\}$	r	$\{A,L\}\{H,M\}$
f	$\{G,L\}\{K,F\}$	s	$\{C,E\}\{D,I\}$
g	$\{B,O\}\{I,J\}$	t	$\{J,L\}\{D,N\}$
h	$\{B,L\}\{H,O\}$	u	$\{B,M\}\{I,N\}$
i	$\{D,M\}\{G,L\}$	v	$\{A,O\}\{N,I\}$
j	$\{E,K\}\{C,L\}$	w	$\{D,N\}\{H,K\}$
k	$\{B,L\}\{F,N\}$	x	$\{D,I\}\{F,G\}$
l	$\{A,C\}\{B,G\}$	y	$\{C,N\}\{J,O\}$
m	$\{E,N\}\{H,M\}$	z	$\{E,G\}\{I,L\}$
n	$\{C,M\}\{G,O\}$	α	$\{E,J\}\{G,H\}$
o	$\{A,C\}\{B,G\}$	β	$\{A,D\}\{E,J\}$
p	$\{G,H\}\{F,J\}$	γ	$\{E,O\}\{F,M\}$
q	$\{C,M\}\{I,K\}$	δ	$\{B,K\}\{J,O\}$
r	$\{A,L\}\{H,M\}$	ϵ	$\{B,K\}\{C,N\}$
s	$\{C,E\}\{D,I\}$	ζ	$\{C,E\}\{F,G\}$
t	$\{J,L\}\{D,N\}$	η	$\{A,L\}\{E,N\}$
u	$\{B,M\}\{I,N\}$	θ	$\{F,M\}\{I,L\}$
v	$\{A,O\}\{N,I\}$	ι	$\{H,K\}\{J,L\}$
w	$\{D,N\}\{H,K\}$	κ	$\{A,C\}\{I,J\}$
x	$\{D,I\}\{F,G\}$	λ	$\{A,D\}\{G,H\}$
y	$\{C,N\}\{J,O\}$	μ	$\{G,O\}\{D,K\}$
z	$\{E,G\}\{I,L\}$	ν	$\{F,N\}\{H,O\}$
α	$\{E,J\}\{G,H\}$	ξ	$\{A,O\}\{B,M\}$
β	$\{A,D\}\{E,J\}$	o	$\{B,E\}\{F,J\}$
γ	$\{E,O\}\{F,M\}$	π	$\{A,M\}\{E,K\}$
δ	$\{B,K\}\{J,O\}$	ρ	$\{C,L\}\{D,O\}$
ϵ	$\{B,K\}\{C,N\}$	σ	$\{O,N\}\{J,M\}$
ζ	$\{C,E\}\{F,G\}$	τ	$\{B,E\}\{H,I\}$

I have constructed on a very large scale a figure of the sixty Pascalian lines, and the forty-five Pascalian points, marking them according to the foregoing notation; but the figure is from its complexity, and the inconvenient way in which the points are either crowded together or fly off to a great distance, almost unintelligible.

[*The two plates (I and II) show the complete table of intersections of the 60 Pascalian lines. Each entry shows the type of point (g, m, h, z, p, r, t, or w) at which any two Pascalian lines meet.*]

402. On a Singularity of Surfaces

[From the Quarterly Journal of Pure and Applied Mathematics, vol. ix. (1868), pp. 332—338.]

A SURFACE having a nodal line has in general on this nodal line points where the two tangent planes coincide, or as I propose to term them "pinch-points." Thus, if the nodal line be the curve of complete intersection of any two surfaces $P = 0$, $Q = 0$, then the equation of the general surface having this curve for a nodal line is

$$(a, b, c \between P, Q)^2 = 0$$

(where a, b, c are any functions of the coordinates), and the pinch-points are given as the intersections of the nodal line $P = 0$, $Q = 0$ with the surface $ac - b^2 = 0$.

Consider the case where the nodal curve is a curve of partial intersection represented by the equations

$$\frac{P}{P'} = \frac{Q}{Q'} = \frac{R}{R'}, \quad \text{or say by the equations } p = 0,\ q = 0,\ r = 0,$$

(viz. p, q, r denote the functions $QR' - Q'R$, $RP' - R'P$, $PQ' - P'Q$ respectively), and consequently we have identically

$$(P, Q, R \between p, q, r)^2 = 0, \quad (P', Q', R' \between p, q, r)^2 = 0,$$

or what is the same thing, (λ, μ) being arbitrary,

$$(\lambda P + \mu P', \lambda Q + \mu Q', \lambda R + \mu R' \between p, q, r)^2 = 0.$$

The general surface having the curve in question for its nodal line is represented by the equation

$$(a, b, c, f, g, h \between p, q, r)^2 = 0,$$

(where (a, b, c, f, g, h) are any functions of the coordinates), and it is easy to see that the condition for a pinch-point is the same as that which (considering p, q, r as coordinates and all the other quantities as constants), expresses that the line

$$(\lambda P + \mu P', \lambda Q + \mu Q', \lambda R + \mu R' \between p, q, r)^2 = 0$$

touches the conic

$$(a, b, c, f, g, h \between p, q, r)^2 = 0,$$

viz. A, B, C, F, G, H being the inverse coefficients, $A = bc - f^2$, &c., this condition is

$$(A, B, C, F, G, H \between \lambda P + \mu P', \lambda Q + \mu Q', \lambda R + \mu R')^2 = 0,$$

or what is the same thing, the pinch-points are given as the common intersections of the nodal line $p = 0$, $q = 0$, $r = 0$ with each of the three surfaces

$$\begin{aligned}(A, B, C, F, G, H \between P, Q, R)^2 &= 0,\\ (A, B, C, F, G, H \between P, Q, R)(P', Q', R') &= 0,\\ (A, B, C, F, G, H \between P', Q', R')^2 &= 0,\end{aligned}$$

these last three equations in fact, adding only a single relation to the relations expressed by the equations $p = 0$, $q = 0$, $r = 0$.

If the functions P, Q, R, P', Q', R' are linear functions of the coordinates, then the nodal curve is a cubic curve in space, the equations $p = 0$, $q = 0$, $r = 0$ represent each of them a quadric surface, and the curve is the partial intersection of any two of these quadric surfaces. The pinch-points are the intersections of the cubic curve with the three quadric surfaces represented by the last-mentioned equations; and since the cubic curve lies on each of the three quadric surfaces $p = 0$, $q = 0$, $r = 0$, the number of pinch-points is at once seen to be $= 12$.